

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Award

Tuesday 24 June 2025

Afternoon (Time: 3 hours)

Paper reference **9811/01**

Advanced Extension Award

Mathematics *Worked Solutions*

You must have:
 Mathematical Formulae and Statistical Tables
 An insert for questions 5, 6 and 7 (enclosed)

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- **Calculators may not be used.**
- You must **show all your working.**
- Answers should be given in as simple a form as possible, e.g. $\frac{2\pi}{3}$, $\sqrt{2}$, $3\sqrt{2}$.

Information

- The total mark for this paper is 100 of which **7** marks are for style and clarity of presentation.
- The style and clarity of presentation marks will be indicated as **(+S1)** or **(+S2)**.
- There are 7 questions in this question paper.
- The marks for each question are shown in brackets.
- The total mark for each question is shown at the end of the question.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P76378A

©2025 Pearson Education Ltd.
Y:1/1/1/




Pearson

1. Using the substitution $u = \frac{\pi}{2}\sqrt{1+x}$ evaluate

$$\int_{\frac{5}{4}}^{\frac{21}{4}} \sec^2\left(\frac{\pi}{2}\sqrt{1+x}\right) dx$$

(10)

$$u = \frac{\pi}{2} (1+x)^{1/2} \quad \left(\frac{2u}{\pi}\right)^2 = (1+x)^{-1/2}$$

$$\frac{du}{dx} = \frac{\pi}{2} \times \frac{1}{2} (1+x)^{-1/2} = \frac{\pi}{4} (1+x)^{-1/2} = \frac{\pi}{4} \times \left(\frac{2u}{\pi}\right)^{-1} = \frac{\pi^2}{8u}$$

$$du = \frac{\pi^2}{8u} dx$$

$$\text{When } x = \frac{21}{4} \quad u = \frac{\pi}{2} \sqrt{\frac{25}{4}} = \frac{5\pi}{4}$$

$$\frac{8u}{\pi^2} du = dx$$

$$\text{When } x = \frac{5}{4} \quad u = \frac{\pi}{2} \sqrt{\frac{9}{4}} = \frac{3\pi}{4}$$

$$\int \sec^2(u) dx = \int \sec^2(u) \cdot \frac{8u}{\pi^2} du = \frac{8}{\pi^2} \int u \sec^2(u) du$$

$$a = u \quad b = \tan(u)$$

$$\frac{da}{du} = 1 \quad \frac{db}{du} = \sec^2(u)$$

$$\frac{8}{\pi^2} \left[u \tan(u) - \int \tan(u) du \right]_{\frac{3\pi}{4}}^{\frac{5\pi}{4}} = \frac{8}{\pi^2} \left[u \tan(u) - \ln|\sec(u)| \right]_{\frac{3\pi}{4}}^{\frac{5\pi}{4}}$$

$$= \frac{8}{\pi^2} \left[\left(\frac{5\pi}{4} \tan\left(\frac{5\pi}{4}\right) - \ln\left|\sec\left(\frac{5\pi}{4}\right)\right| \right) - \left(\frac{3\pi}{4} \tan\left(\frac{3\pi}{4}\right) - \ln\left|\sec\left(\frac{3\pi}{4}\right)\right| \right) \right]$$

$$= \frac{8}{\pi^2} \left[\frac{5\pi}{4} - \ln|-\sqrt{2}| - \left(-\frac{3\pi}{4} + \ln|-\sqrt{2}| \right) \right]$$

$$\frac{8}{\pi^2} \left[\frac{8\pi}{4} \right] = \frac{16}{\pi}$$



Question 1 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 1 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 1 continued

(Total for Question 1 is 10 marks)

2. (a) On separate diagrams sketch the graph of

(i) $y = |x^2 - 4|$ (3)

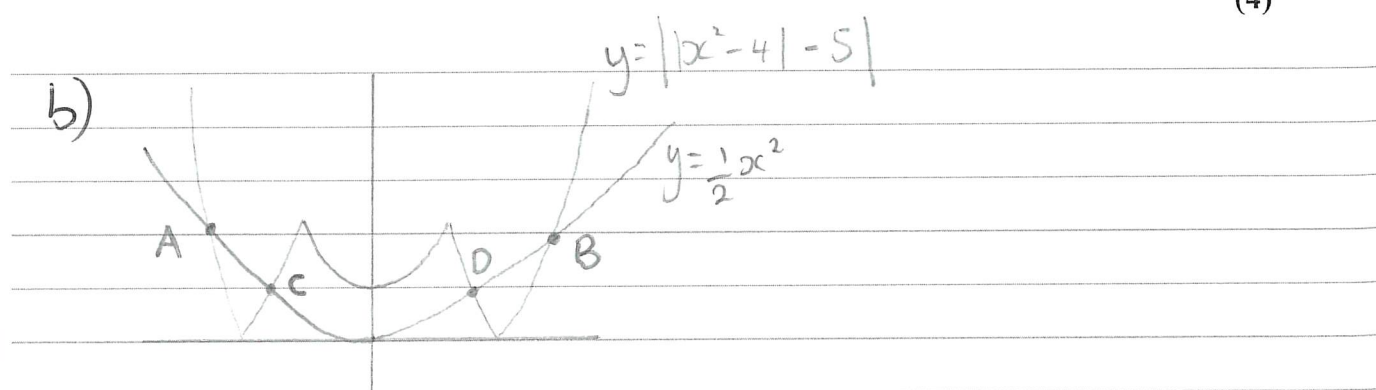
(ii) $y = ||x^2 - 4| - 5|$ (4)

Indicate clearly on each sketch the coordinates of

- any vertices
- any points where the graph touches or crosses the coordinate axes

(b) Hence solve the equation

$$||x^2 - 4| - 5| = \frac{1}{2}x^2 \quad (4)$$



Points A and B are when $\frac{1}{2}x^2$ intersect the original x^2 translated

$$x^2 - 4 - 5 = \frac{1}{2}x^2$$

$$x^2 - 9 = \frac{1}{2}x^2$$

$$\frac{1}{2}x^2 = 9$$

$$x^2 = 18$$

$$x = \pm\sqrt{18} = \pm\sqrt{9}\sqrt{2} = \pm 3\sqrt{2}$$

Points C and D are when $\frac{1}{2}x^2$ intersect the reflected $x^2 - 4$ translated

$$-[(x^2 - 4) - 5] = \frac{1}{2}x^2$$

$$-x^2 + 9 = \frac{1}{2}x^2$$

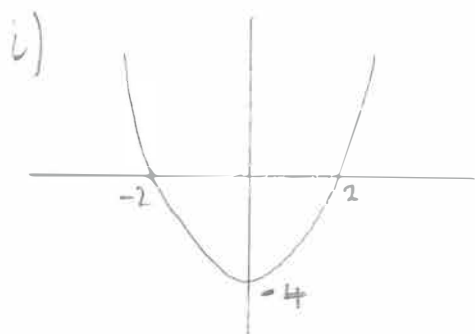
$$9 = \frac{3}{2}x^2$$

$$6 = x^2$$

$$x = \pm\sqrt{6}$$



Question 2 continued

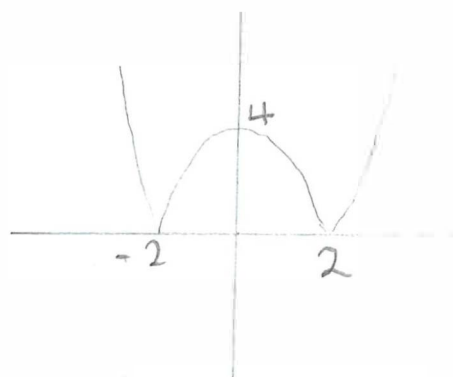


$$y = x^2 - 4$$

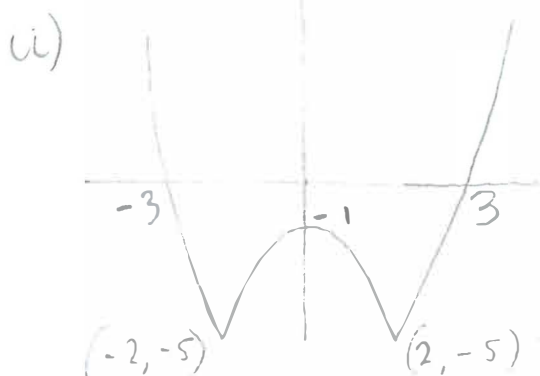
$$0 = x^2 - 4$$

$$4 = x^2$$

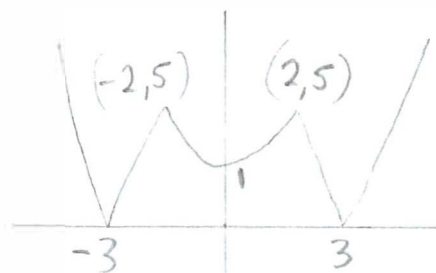
$$\pm 2 = x$$



$$y = |x^2 - 4|$$



$$y = |x^2 - 4| - 5$$



$$y = ||x^2 - 4| - 5|$$

Roots on original curve
not reflected so

$$0 = (x^2 - 4) - 5$$

$$0 = x^2 - 9$$

$$9 = x^2$$

$$\pm 3 = x$$

Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 2 is 11 marks)



3. Solve the equation

$$\arcsin 2x = \arccos x - \frac{\pi}{6} \quad 0 \leq x \leq 0.5$$

(5)

$$\sin(\arcsin(2x)) = \sin(\arccos(x) - \pi/6)$$

$$2x = \sin \arccos(x) \cos(\pi/6) - \sin(\pi/6) \cos \arccos(x)$$

$$2x = \sin \arccos(x) \cdot \frac{\sqrt{3}}{2} - \frac{1}{2}x$$

$$\frac{5x}{2} = \frac{\sqrt{3}}{2} \sin \arccos(x)$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$\frac{5x}{\sqrt{3}} = \sin \arccos(x)$$

$$\theta = \arccos(x)$$

$$\sin \arccos(x) = \sqrt{1 - (\cos \arccos(x))^2} = \sqrt{1 - x^2}$$

$$\left(\frac{5x}{\sqrt{3}}\right)^2 = \left(\sqrt{1 - x^2}\right)^2$$

$$\frac{25x^2}{3} = 1 - x^2$$

$$\frac{28x^2}{3} = 1$$

$$x^2 = \frac{3}{28}$$

$$x = \frac{\sqrt{3}}{\sqrt{28}} = \frac{\sqrt{3}}{\sqrt{28}} = \frac{\sqrt{3}}{2\sqrt{7}} = \frac{\sqrt{3} \times \sqrt{7}}{2\sqrt{7} \times \sqrt{7}} = \frac{\sqrt{21}}{14}$$



Question 3 continued

(Total for Question 3 is 5 marks)

4. In this question you may assume that $\sqrt{p}$ is irrational for any prime number p .

- (a) Use proof by contradiction to show that there is no rational number a and integer b for which

$$\sqrt{5} = a + b\sqrt{7} \quad (5)$$

- (b) Hence prove that $\sqrt{5} + \sqrt{7}$ is not a rational number.

(2)
(+S1)

Assume there is a rational number a and integer b such that $\sqrt{5} = a + b\sqrt{7}$

$$(\sqrt{5} - a)^2 = (b\sqrt{7})^2$$

$$5 - 2a\sqrt{5} + a^2 = 7b^2$$

$$5 + a^2 - 7b^2 = 2a\sqrt{5}$$

If a is rational a^2 is rational

If b is an integer b^2 is rational

So $a^2 - 7b^2 + 5$ is rational

This is a contradiction as $2a\sqrt{5}$ is irrational unless $a = 0$

If $a = 0$ $5 = 7b^2$ so $5/7 = b^2$ which is not possible as b is an integer so $a \neq 0$ therefore contradiction.

Hence there is no rational a and integer b for which $\sqrt{5} = a + b\sqrt{7}$

b) If $b = -1$ then from (a) $\sqrt{5} = a - \sqrt{7}$ and a must be irrational.

$\sqrt{5} + \sqrt{7} = a$
therefore $\sqrt{5} + \sqrt{7}$ is irrational



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 8 marks)



5. The discrete random variable X has the probability distribution

n	2	4	6	8	10
$P(X=n)$	$\frac{1}{10} \operatorname{cosec}^2 x$	a	$\frac{1}{5} \cot^2 x$	$\frac{1}{10} \sec^2 x$	$\frac{1}{5} \tan^2 x$

where a and x are constants.

(a) Show that

$$\sin^2 2x = \frac{6}{7-5a} \quad (7)$$

(b) Use the result from part (a) to find a restriction on the maximum possible value of a . (2)

The random variables X_1 and X_2 are independent and each have the same distribution as X .

Given that $P(X_1 \times X_2 = 16) = a - \frac{1}{25}$

(c) determine any possible values of a , justifying your answer.

(8)
(+S2)

$$\sum P(X=n) = 1$$

$$\frac{1}{10} \operatorname{cosec}^2(x) + a + \frac{1}{5} \cot^2(x) + \frac{1}{10} \sec^2(x) + \frac{1}{5} \tan^2(x) = 1$$

$$\textcircled{\times 10} \quad \frac{1}{\sin^2(x)} + 10a + 2 \frac{\cos^2(x)}{\sin^2(x)} + \frac{1}{\cos^2(x)} + 2 \frac{\sin^2(x)}{\cos^2(x)} = 10$$

$$\frac{1 + 2 \cos^2(x)}{\sin^2(x)} + \frac{2 \sin^2(x) + 1}{\cos^2(x)} = 10 - 10a$$

$$\textcircled{\times \sin^2(x) \cos^2(x)} \quad (1 + 2 \cos^2(x)) \cos^2(x) + (2 \sin^2(x) + 1) \sin^2(x) = (10 - 10a) \cos^2(x) \sin^2(x)$$

$$\cos^2(x) + 2 \cos^4(x) + 2 \sin^4(x) + \sin^2(x) = (10 - 10a) \cos^2(x) \sin^2(x)$$

$$2(\cos^4(x) + \sin^4(x)) = 2[(\cos^2(x) + \sin^2(x))^2 - 2 \sin^2(x) \cos^2(x)]$$

$$\cos^2(x) + \sin^2(x) = 1 \quad = 2[1 - 2 \sin^2(x) \cos^2(x)] = 2 - 4 \sin^2(x) \cos^2(x)$$

$$\sin(2x) = 2 \sin(x) \cos(x) \quad \sin^2(2x) = 4 \sin^2(x) \cos^2(x)$$



Question 5 continued

$$\cos^2(x) + \sin^2(x) + 2(\cos^4(x) + \sin^4(x)) = (10 - 10a)\cos^2(x)\sin^2(x)$$

$$1 + 2 - 4\sin^2(x)\cos^2(x) = \left(\frac{10 - 10a}{4}\right) \cdot 4\cos^2(x)\sin^2(x)$$

$$3 - \sin^2(2x) = \left(\frac{10 - 10a}{4}\right) \sin^2(2x)$$

$$(\times 4) \quad 12 - 4\sin^2(2x) = (10 - 10a)\sin^2(2x)$$

$$+ 4\sin^2(2x)$$

$$12 = (14 - 10a)\sin^2(2x)$$

$$\frac{12}{14 - 10a} = \frac{6}{7 - 5a} = \sin^2(2x)$$

b) Max $\sin^2(2x) = 1$

$$\frac{6}{7 - 5a} = 1$$

$$6 = 7 - 5a$$

$$5a = 1$$

$$a = 1/5$$

c) Factor pairs of 16

1, 16

2, 8 ✓

4, 4 ✓

8, 2 ✓

16, 1

$$2 \times \frac{1}{10} \operatorname{cosec}^2(x) \times \frac{1}{10} \sec^2(x) + a^2 = a - \frac{1}{25}$$

$$\frac{2}{100 \sin^2(x) \cos^2(x)} + a^2 = a - \frac{1}{25}$$

$$\frac{2}{25 \sin^2(2x)} + a^2 = a - \frac{1}{25}$$

$$\frac{2}{\sin^2(2x)} + 25a^2 = 25a - 1$$

(x25)

$$\frac{2}{6} = 2 \times \frac{7 - 5a}{6} = \frac{7 - 5a}{3}$$

$$7 - 5a$$

$$\frac{7 - 5a}{3} + 25a^2 = 25a - 1$$

(x3)

$$7 - 5a + 75a^2 = 75a - 3$$

$$75a^2 - 80a + 10 = 0$$

(x5)

$$15a^2 - 16a + 2 = 0$$

Question 5 continued

$$a = \frac{16 \pm \sqrt{16^2 - 4 \times 15 \times 2}}{2 \times 15} = \frac{16 \pm \sqrt{256 - 120}}{30}$$

$$a = \frac{16 \pm \sqrt{136}}{30} = \frac{16 \pm 2\sqrt{34}}{30} = \frac{8 \pm \sqrt{34}}{15}$$

$$\begin{array}{ll} \sqrt{34} > \sqrt{25} & \sqrt{34} > 5 \\ \sqrt{34} < \sqrt{36} & \sqrt{34} < 6 \end{array}$$

$$\frac{8 + \sqrt{34}}{15} > \frac{8 + 5}{15} = \frac{13}{15} > \frac{3}{15} = \frac{1}{5} \quad \text{We know from (b) max } a = \frac{1}{5} \text{ not possible.}$$

$$\frac{8 - \sqrt{34}}{15} < \frac{8 - 6}{15} = \frac{2}{15} < \frac{3}{15} = \frac{1}{5}$$

$$\text{So } a = \frac{8 - \sqrt{34}}{15}$$



Question 5 continued

(Total for Question 5 is 19 marks)

6. Relative to a fixed origin O , the points A , B and C have position vectors

$$\vec{OA} = 10\mathbf{i} + 5\mathbf{j} + 4\mathbf{k} \quad \vec{OB} = -4\mathbf{i} + 7\mathbf{j} - \mathbf{k} \quad \vec{OC} = -2\mathbf{i} + 17\mathbf{j} + 10\mathbf{k}$$

(a) Show that triangle ABC is isosceles.

(2)

The circle S lies inside triangle ABC such that each side of the triangle is a tangent to S .

The centre, X , of S lies on the angle bisector of each vertex of the triangle.

The point D lies on AC such that $|\vec{AD}| = |\vec{AB}|$

(b) Find the position vector of D .

(2)

(c) Find the position vector of the midpoint of BD .

(2)

(d) Hence show that the position of X is given by

$$\vec{OX} = 10\mathbf{i} + 5\mathbf{j} + 4\mathbf{k} + p(-2\mathbf{i} + \mathbf{j})$$

where p is a scalar constant.

(4)

(e) Use the angle bisector of vertex B to find another expression for $\vec{OX}$ in terms of a scalar constant q .

(3)

(f) By equating the $\mathbf{k}$ components of the two expressions for $\vec{OX}$, determine the value of q .

(2)

(g) Find

(i) the position vector of the centre of S ,

(2)

(ii) the radius of S .

(4)

$$\begin{aligned} \text{a) } \vec{AB} &= \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} \rightarrow \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} = \begin{pmatrix} -14 \\ 2 \\ -5 \end{pmatrix} & |\vec{AB}| &= \sqrt{14^2 + 2^2 + 5^2} \\ & & & = \sqrt{225} \\ & & & = 15 \\ \vec{BC} &= \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} \rightarrow \begin{pmatrix} -2 \\ 17 \\ 10 \end{pmatrix} = \begin{pmatrix} 2 \\ 10 \\ 11 \end{pmatrix} & |\vec{BC}| &= \sqrt{2^2 + 10^2 + 11^2} \\ & & & = \sqrt{225} \\ & & & = 15 \\ \vec{AC} &= \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} \rightarrow \begin{pmatrix} -2 \\ 17 \\ 10 \end{pmatrix} = \begin{pmatrix} -12 \\ 12 \\ 6 \end{pmatrix} & |\vec{AC}| &= \sqrt{12^2 + 12^2 + 6^2} \\ & & & = \sqrt{324} \\ & & & = 18 \end{aligned}$$

$|\vec{AB}| = |\vec{AC}| \quad \therefore \text{isosceles}$



Question 6 continued

$$b) \hat{AC} = \frac{1}{18} \begin{pmatrix} -12 \\ 12 \\ 6 \end{pmatrix}$$

$$\vec{AD} = 15 \hat{AC}$$

$$\begin{aligned} \vec{OD} &= \vec{OA} + \vec{AD} = \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} + 15 \left(\frac{1}{18} \begin{pmatrix} -12 \\ 12 \\ 6 \end{pmatrix} \right) \\ &= \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} + \begin{pmatrix} -10 \\ 10 \\ 5 \end{pmatrix} = \begin{pmatrix} 0 \\ 15 \\ 9 \end{pmatrix} \end{aligned}$$

$$c) \vec{BD} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 15 \\ 9 \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \\ 10 \end{pmatrix} \quad \frac{1}{2} \vec{BD} = \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix}$$

Call the midpoint of BD, M.

$$\vec{OM} = \vec{OB} + \frac{1}{2} \vec{BD} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} -2 \\ 11 \\ 4 \end{pmatrix}$$

d) ABD is isosceles so the bisector of angle A goes through the midpoint of BD, M.

$$\vec{AM} = \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} \rightarrow \begin{pmatrix} -2 \\ 11 \\ 4 \end{pmatrix} = \begin{pmatrix} -12 \\ 6 \\ 0 \end{pmatrix}$$

X lies on the bisector of angle A. $\vec{AX} = \lambda \begin{pmatrix} -12 \\ 6 \\ 0 \end{pmatrix}$

$$\vec{OX} = \vec{OA} + \vec{AX} = \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} -12 \\ 6 \\ 0 \end{pmatrix} = \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} + 6\lambda \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

$$\vec{OX} = 10\mathbf{i} + 5\mathbf{j} + 4\mathbf{k} + p(-2\mathbf{i} + \mathbf{j}) \quad \text{where } p = 6\lambda$$

Question 6 continued

e) ABC is isosceles so the bisector of B goes through the midpoint of AC, say N.

$$\vec{ON} = \vec{OA} + \frac{1}{2} \vec{AC} = \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} -12 \\ 12 \\ 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 11 \\ 7 \end{pmatrix}$$

X lies on the bisector of B

$$\vec{BN} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} \rightarrow \begin{pmatrix} 4 \\ 11 \\ 7 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \\ 8 \end{pmatrix} \quad \vec{BX} = \mu \begin{pmatrix} 8 \\ 4 \\ 8 \end{pmatrix}$$

$$\vec{OX} = \vec{OB} + \vec{BX} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 8 \\ 4 \\ 8 \end{pmatrix} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} + 4\mu \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$$

$$\vec{OX} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} + q \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \quad \text{where } q = 4\mu$$

f) k components can be equated: $4 = -1 + 2q$
 $\frac{5}{2} = q$

gi) $\vec{OX} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} + \frac{5}{2} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 9.5 \\ 4 \end{pmatrix}$

gii) N lies on the circle, hence $|\vec{XN}| = r$

$$\vec{XN} = \begin{pmatrix} 1 \\ 9.5 \\ 4 \end{pmatrix} \rightarrow \begin{pmatrix} 4 \\ 11 \\ 7 \end{pmatrix} = \begin{pmatrix} 3 \\ 1.5 \\ 3 \end{pmatrix} \quad |\vec{XN}| = \sqrt{3^2 + 1.5^2 + 3^2}$$

$$= \sqrt{18 + \frac{9}{4}} = \sqrt{\frac{81}{4}}$$

$$r = \frac{9}{2}$$



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 23 marks)



7. Given $u = y^x$

(a) by taking $\ln$ of both sides, or otherwise, determine $\frac{du}{dx}$ in terms of x , y and $\frac{dy}{dx}$ only.

(5)

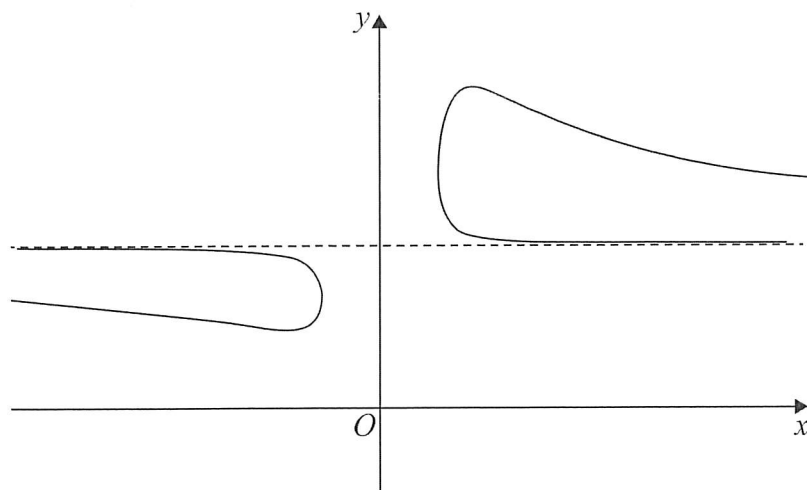


Figure 1

Figure 1 shows a sketch of the curve given by the equation

$$3y^x - x^3 \ln y = 0$$

The curve has a horizontal asymptote $y = 1$, and is defined for

- $x \leq -a, \quad x \geq a$
- $b \leq y \leq c \quad y \neq 1$

where a , b and c are positive constants.

(b) Determine $\frac{dy}{dx}$ for this curve.

(4)

(c) Find the value of a .

(6)

(d) Find the value of b and the value of c .

(7)

(+S2)

a) $\ln u = x \ln y$

RHS $a = x \quad b = \ln y$

$$\frac{da}{dx} = 1 \quad \frac{db}{dx} = \frac{1}{y} \frac{dy}{dx}$$

$$\frac{1}{u} \frac{du}{dx} = \ln y + \frac{x}{y} \frac{dy}{dx}$$

$$b \frac{da}{dx} + a \frac{db}{dx}$$

$$\frac{du}{dx} = \left(\ln y + \frac{x}{y} \frac{dy}{dx} \right) u = \left(\ln y + \frac{x}{y} \frac{dy}{dx} \right) y^x$$



Question 7 continued

$$b) 3y^x = x^3 \ln y$$

RHS

$$a = x^3$$

$$b = \ln y$$

$$\frac{da}{dx} = 3x^2$$

$$\frac{db}{dx} = \frac{1}{y} \frac{dy}{dx}$$

$$3 \left[\left(\frac{x}{y} \frac{dy}{dx} + \ln y \right) y^x \right] = 3x^2 \ln y + \frac{x^3}{y} \frac{dy}{dx}$$

$$3 \frac{x y^x}{y} \frac{dy}{dx} + 3 y^x \ln y = 3x^2 \ln y + \frac{x^3}{y} \frac{dy}{dx}$$

$$\frac{3 x y^x}{y} \frac{dy}{dx} - \frac{x^3}{y} \frac{dy}{dx} = 3x^2 \ln y - 3 y^x \ln y$$

$$\frac{dy}{dx} \left(\frac{3 x y^x}{y} - \frac{x^3}{y} \right) = 3x^2 \ln y - 3 y^x \ln y$$

$$\frac{dy}{dx} = (3x^2 \ln y - 3 y^x \ln y) \times \frac{y}{3 x y^x - x^3} = \frac{3 y \ln y (x^2 - y^x)}{3 x y^x - x^3}$$

c) at $x=a$ the tangent is vertical so $\frac{dy}{dx} = \infty$

$$\Rightarrow 3x y^x - x^3 = 0$$

$$x(3y^x - x^2) = 0$$

$x \neq 0$ from graph

$$3y^x = x^2$$

$$y^x = \frac{1}{3} x^2 \quad (1)$$

$$x \ln y = \ln \left(\frac{x^2}{3} \right) \quad (2)$$

Using the eq of the curve $3y^x - x^3 \ln y = 0$ with result (1) gives:

$$3 \left(\frac{1}{3} x^2 \right) - x^3 \ln y = 0$$

$$x^2 (1 - x \ln y) = 0 \quad x^2 \neq 0$$

$$1 = x \ln y$$

Using result (2) above

$$1 = \ln \left(\frac{x^2}{3} \right)$$

$$e^1 = \frac{x^2}{3}$$

$$3e = x^2 \quad x = \pm \sqrt{3e}$$

$$a = \sqrt{3e}$$

Question 7 continued

d) at $y=b$ and $y=c$ the tangent is horizontal so $\frac{dy}{dx} = 0$

$$\Rightarrow 3y \ln y (x^2 - y^x) = 0 \quad \ln y \neq 0 \text{ since } y \neq 1$$

$$x^2 = y^x$$

$$\ln x^2 = x \ln y$$

Using eq of curve $3y^x - x^3 \ln y = 0$ with this result gives

$$3x^2 - x^3 \ln y = 0$$

$$3x^2 - x^3 x \ln y = 0$$

$$3x^2 - x^2 \ln x^2 = 0$$

$$x^2 (3 - \ln x^2) = 0 \quad x^2 \neq 0$$

$$3 = \ln x^2$$

$$e^3 = x^2$$

$$\pm \sqrt{e^3} = \pm (e^3)^{1/2} = \pm e^{3/2} = x$$

from $\ln x^2 = x \ln y$ we have $3 = x \ln y$

$$\text{when } x = e^{3/2}$$

$$\frac{3}{x} = \ln y \quad y = e^{3/x}$$

$$y = e^{3/e^{3/2}} = e^{3e^{-3/2}}$$

$$\text{so } c = e^{3e^{-3/2}}$$

$$\text{when } x = -e^{3/2}$$

$$y = e^{3/(-e^{3/2})} = e^{-3e^{-3/2}}$$

$$\text{so } b = e^{-3e^{-3/2}}$$



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 7 is 24 marks)

TOTAL FOR PAPER IS 100 MARKS



Pearson Edexcel Award

Tuesday 24 June 2025

Afternoon (Time: 3 hours)

Paper
reference

9811/01

**Advanced Extension Award
Mathematics**

Insert for questions 5, 6 and 7
Do not write on this insert.

Turn over ►

P76378A

©2025 Pearson Education Ltd.
Y:1/1/1/



5. The discrete random variable X has the probability distribution

n	2	4	6	8	10
$P(X = n)$	$\frac{1}{10} \operatorname{cosec}^2 x$	a	$\frac{1}{5} \cot^2 x$	$\frac{1}{10} \sec^2 x$	$\frac{1}{5} \tan^2 x$

where a and x are constants.

(a) Show that

$$\sin^2 2x = \frac{6}{7 - 5a} \quad (7)$$

(b) Use the result from part (a) to find a restriction on the maximum possible value of a . (2)

The random variables X_1 and X_2 are independent and each have the same distribution as X .

Given that $P(X_1 \times X_2 = 16) = a - \frac{1}{25}$

(c) determine any possible values of a , justifying your answer. (8)
(+S2)

(Total for Question 5 is 19 marks)

6. Relative to a fixed origin O , the points A , B and C have position vectors

$$\vec{OA} = 10\mathbf{i} + 5\mathbf{j} + 4\mathbf{k} \quad \vec{OB} = -4\mathbf{i} + 7\mathbf{j} - \mathbf{k} \quad \vec{OC} = -2\mathbf{i} + 17\mathbf{j} + 10\mathbf{k}$$

(a) Show that triangle ABC is isosceles.

(2)

The circle S lies inside triangle ABC such that each side of the triangle is a tangent to S .

The centre, X , of S lies on the angle bisector of each vertex of the triangle.

The point D lies on AC such that $|\vec{AD}| = |\vec{AB}|$

(b) Find the position vector of D .

(2)

(c) Find the position vector of the midpoint of BD .

(2)

(d) Hence show that the position of X is given by

$$\vec{OX} = 10\mathbf{i} + 5\mathbf{j} + 4\mathbf{k} + p(-2\mathbf{i} + \mathbf{j})$$

where p is a scalar constant.

(4)

(e) Use the angle bisector of vertex B to find another expression for $\vec{OX}$ in terms of a scalar constant q .

(3)

(f) By equating the $\mathbf{k}$ components of the two expressions for $\vec{OX}$, determine the value of q .

(2)

(g) Find

(i) the position vector of the centre of S ,

(2)

(ii) the radius of S .

(4)

(+S2)

(Total for Question 6 is 23 marks)

7. Given $u = y^x$

- (a) by taking $\ln$ of both sides, or otherwise, determine $\frac{du}{dx}$ in terms of x , y and $\frac{dy}{dx}$ only. (5)

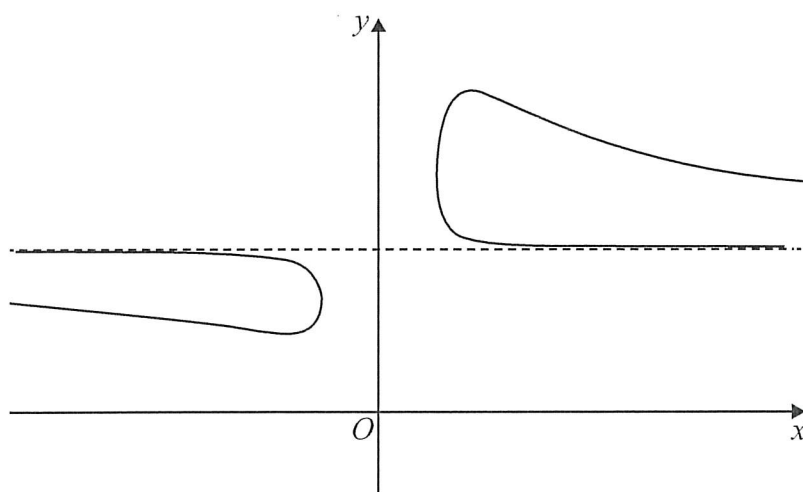


Figure 1

Figure 1 shows a sketch of the curve given by the equation

$$3y^x - x^3 \ln y = 0$$

The curve has a horizontal asymptote $y = 1$, and is defined for

- $x \leq -a, \quad x \geq a$
- $b \leq y \leq c \quad y \neq 1$

where a , b and c are positive constants.

- (b) Determine $\frac{dy}{dx}$ for this curve. (4)
- (c) Find the value of a . (6)
- (d) Find the value of b and the value of c . (7)

(+S2)

(Total for Question 7 is 24 marks)